

Slides Outline — Submitting AI Jobs to the Anvil Supercomputer

~18 slides for the 15-min framing + live-demo anchors + 15-min wrap-up. Build in whichever deck tool you prefer; each slide below is one screen of content.

Framing (1:00–1:15 PM)

#	Slide title	Content bullets	Anchor
1	Title	“Submitting AI Jobs to the Anvil Supercomputer” ; AI for All, Beginner Track 1 ; Sept 15 2026 ; your name/affiliation (ASU RTO)	—
2	Who this is for	No HPC experience needed ; laptop + SSH/browser ; by 2:30 you’ll have run a job on a national supercomputer	set expectations
3	Why a supercomputer for AI?	Your laptop vs. an A100/H100 ; training/inference scale ; connects to this morning’s “Leveraging National Level AI and HPC Resources” talk	bridge from morning
4	Meet Anvil	Purdue ; NSF-funded ACCESS resource ; 1,000+ nodes ; 128 cores/node AMD EPYC ; 16x A100 nodes, 21x H100 nodes ; 5.3 PF	verified specs
5	Why queues?	Shared machine, fair share ; login node is not compute node ; running on the login node is against Anvil policy	say twice
6	Slurm in one slide	sbatch / squeue / scancel / sacct ; job = a shell script + a resource request ; the batch model: submit -> queue -> run -> output file	—
7	Anvil’s queues (partitions)	Table: debug / gpu-debug / shared / wholenode / wide / highmem / gpu / ai with limits ; debug queues are free-ish and short = perfect for a class	showpartitions demo screenshot

#	Slide title	Content bullets	Anchor
8	Anatomy of a job script	Annotated hello.sbatch (the actual lab file) ; the two mandatory lines: -A account, -p partition ; resource lines: nodes, tasks, cpus, mem, time, gres	—
9	Today's roadmap	Lab 1 connect -> Lab 2 first job -> Lab 3 real AI job (CPU) -> Lab 4 GPU job -> stretch: job arrays	checkpoints marked

Live-demo anchors during labs (instructor screen)

#	Slide title	Content bullets	Anchor
10	Lab 1 — Connect & look around	ssh command ; OnDemand browser fallback URL ; mybalance / showpartitions / sfeatures output (screenshot from pre-session run)	live typing
11	Lab 2 — Your first job	sbatch -> job ID -> queue -> cat output file ; highlight SLURM_JOB_NODELIST: it ran on a different machine	live typing
12	Lab 3 — A real AI job	train_digits.py idea (no download) ; cpus-per-task=16 ; reading digits_metrics.json ; “what did this cost in SUs?”	live typing
13	Lab 4 — Your first GPU job	—gres=gpu:1 ; gpu-debug partition (30 min, 2 GPUs) ; nvidia-smi in the output = the wow moment ; NGC containers mention	live typing
14	Stretch — Job arrays	—array=0-4 ; SLURM_ARRAY_TASK_ID ; one script, five experiments	if time

Wrap-up (2:15–2:30 PM)

#	Slide title	Content bullets	Anchor
15	What you did today	connected; submitted; monitored; trained on CPU; trained on an A100	checklist
16	Getting your own allocation	ACCESS credits -> transfer to Anvil (allocations.access-ci.org) ; NAIRR Pilot: allocation automatic on award (nairrpilot.org) ; ask your institution's research computing office	next steps
17	Rules of the road	Login-node etiquette ; debug queues for tests ; charge factors (highmem 4x; wholenode bills 128 cores even if you use 1) ; where to get help (rcac-help@purdue.edu, docs)	—
18	Q&A + QR code	QR -> lab files + docs link ; stay through the break — Beginner Track 2 (API to a model) starts 2:45	close